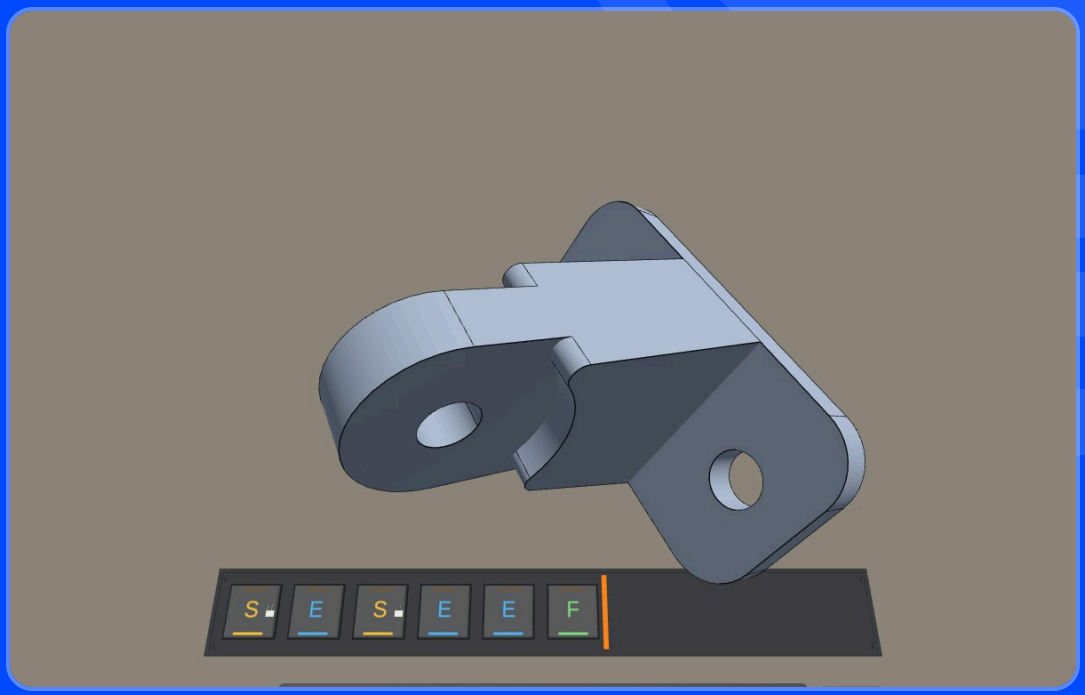
SpaceSuit CAD

QUICK START GUIDE

From empty room to finished part.

An introduction to SpaceSuit CAD, and a bracket built from its drawing, step by step.



BUILD
11 October 2026

DEVICE
Meta Quest 3 · controllers or hands

PUBLISHER
SpacesuitXR

1 Meet SpaceSuit CAD

Parametric CAD, in the room where the work is.

SpaceSuit CAD is a parametric, history-based CAD application that runs on its own on a Meta Quest 3. You model the way you would in Fusion or SolidWorks: sketch a profile, dimension and constrain it, then turn it into a solid and refine it with features. The difference is where you do it. The part floats in front of you, in your own room, at any size you like, including its real size.

Everything you make is recorded on a **timeline**. Change a dimension in the first sketch and every feature after it rebuilds. Select any step and press Edit to change how it was made.

What you can do

- **Sketch** lines, rectangles, circles, arcs, slots, polygons, ellipses, splines and points. Add dimensions and relations, then trim, offset, mirror and pattern.
- **Build parts** with extrude and cut, revolve, loft, sweep, rib, holes (standard sizes), fillets and chamfers, shell, draft, patterns, mirror, move, and reference planes and axes.
- **Assemble** saved parts with mates (coincident, concentric, distance, angle, parallel), then drag the mechanism by hand.
- **Vary** a part with configurations.
- **Exchange files**: import STEP parts and PDF or image drawings, export STEP and STL.
- **Check** by measuring distances, diameters and volume, and see the part at 1:1 in passthrough.

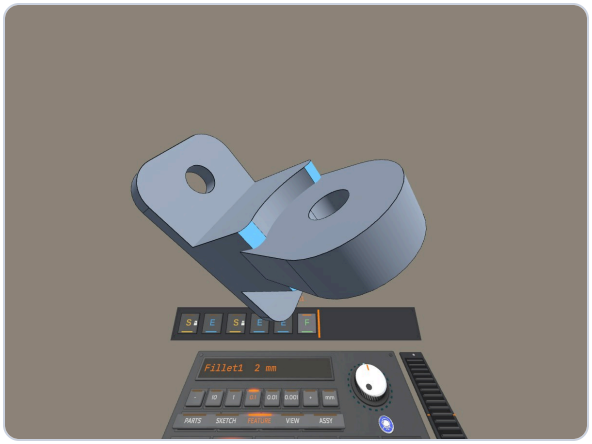


Fig. 1.1 A bracket, being filleted. Every picture in this guide is a rendering of the app itself.

The five modes

The console has a mode key for each kind of work. Each mode shows only its own keys.

Mode	For	Main keys
PARTS	Files	New, Open, Copy, Config, Material, Import, Export
SKETCH	2D profiles	DRAW, CONSTRAIN and EDIT tabs; Finish sketch
FEATURE	The solid	CREATE (Extrude, Revolve, Hole…), MODIFY (Edit, Fillet, Shell…), CONSTRUCT (planes, axes)
VIEW	Looking and checking	Measure, Calc, 1:1, Lock, Room (passthrough), About
ASSY	Assemblies	New, Open, Insert, Fix, Mate, Edit part

Console keys are shown like Extrude, a lit key like Extrude, modes like FEATURE, and readout text like `Extrude1 5 mm join`.

2 Your workspace and controls

Seated, with everything within arm's reach.



Fig. 2.1 The app as it opens: the timeline strip above the console, the dial, the mode keys, and the jog wheel on the right. Passthrough shows your room behind it.

1. Sit at a table and open **SpaceSuit CAD**. The panels place themselves in front of you.
2. Your part floats above the console. While you sketch, a **drafting board** opens on the right.
3. A **keypad** appears whenever a number is wanted. Move it by its handle and it stays where you put it.
4. Your work is saved automatically, including its undo history. The app reopens where you left off.

Controls

SpaceSuit CAD works with the Touch controllers, or with your hands alone. With controllers, point the ray and pull the trigger; with hands, point and pinch. Nearby keys can also be pressed with a fingertip or the tip of a controller.

To	Controller	Hands
Press a key, pick a face, edge or plane	Point, pull the **trigger**	Point and pinch, or touch the key
Draw on the drafting board	Point at the board, trigger	Touch the board with a fingertip
Move and turn the part	**Grip** anywhere; two grips also resize	Pinch the part; two hands also resize
Move a panel	Trigger on its handle, then carry	Pinch its handle, then carry
Edit the selected timeline entry	**B** (or the Edit key)	The Edit key
Give up the current pick	Left trigger	Press the lit key again
Undo	Left thumbstick click	The Undo key
Passthrough on or off	**A**	VIEW › Room

REST THE RAY ON A KEY

Hold the ray (or a finger) still on any key for half a second and a one-line tip tells you what it does.

3 The console and the drafting board

Where you choose what to do, and where you draw.



Fig. 3.1 The console in FEATURE › CREATE. From the top: the readout and the dial; the jog row with the increments (10, 1, 0.1, 0.01, 0.001) and the unit; the five mode keys; the tab keys; the tools.



Fig. 3.2 A key with a small ▾ holds a group. Pinch the ▾ corner to see its variants (here Extrude, Cut, Plane cut and Back). The one you choose stays on the key.



Fig. 3.3 MODIFY with the ray resting on Fillet: its tip reads *Fillet: pinch the edges, type the radius.*

Setting sizes

Every size (an extrude's depth, a fillet's radius, a dimension) can be set four ways, all acting on what is selected:

- **Arrow:** features like Extrude show an amber arrow on the part. Pull it for a rough size in 10 mm steps.
- **Keypad:** opens on the exact value. It also does arithmetic (/ x - +) and has an in/mm key.

- **Dial and jog wheel:** each click is one step of the lit increment. Choose the increment on the jog row, and turn for fine adjustment.
- − and +: one step each.

The drafting board

When a sketch opens, the part glides onto the board so the sketch plane lies flat in front of you; anything in front of the plane is sliced away. Draw with a fingertip or the ray. The board's top rim has keys to pan (< ^ v >), zoom (− +) and turn the view (↻).



Fig. 3.4 The DRAW tab. Upper row: DRAW, CONSTRAIN, EDIT, Undo, Cancel and the green Finish sketch. Below them, one key per kind of shape.



Fig. 3.5 The Polygon ▾ open: polygons by their number of sides, and versions sized across the flats.

Tab	Keys (▾ marks a group)
DRAW	Line ▾ (Centerline) · Rect ▾ (C.Rect, 3pt Rect) · Circle ▾ (Ellipse) · Arc ▾ (3pt Arc, T.Arc) · Slot ▾ (Arc slot) · Polygon ▾ · Spline · Point · Dimension · Constr
CONSTRAIN	Dimension · Horiz ▾ Vert · Coinc ▾ Mid · Symm · Parallel · Perp · Tangent · Equal · Fix
EDIT	Delete · Project · Trim ▾ Extend · Fillet · Offset · Mirror · Rotate ▾ (Copy, Row, Around) · Plane ▾ (Plane offset)

Drawing

- Choose a tool and touch the points it asks for. A message above the board says what comes next.
- **Type a size while drawing:** once a rectangle, circle, slot or polygon is started, the keypad asks for its size. Type it, then pinch to place the shape.
- Lines snap to horizontal and vertical, and points snap to ends, middles, centres and the part's own edges.
- Touch a line's last point again to stop it, or its first point to close it.

Dimensions and relations

- Dimension: pick a line for its length, a circle for its radius, or two lines for their angle, then type the value.
- On a part's face, Dimension and the relations can pick the face's sides directly.
- The sketch turns from blue to white as it becomes fully constrained.
- With no tool chosen, tap to select and drag to move.

4 Quick start: a bracket

Practice Problem 1.6, built from its drawing in ten steps.

This chapter builds a complete part from a dimensioned drawing, the way you would at a desk: a 60 × 30 × 5 mm plate with two Ø8 holes, and an arm that rises at 15° to a round end with a Ø8 hole, its lug 9 mm wide and its block 18 mm wide. Read every size off the drawing; the steps say where each one goes. On the headset, import the drawing's PDF with PARTS › Import to keep it beside you while you work.

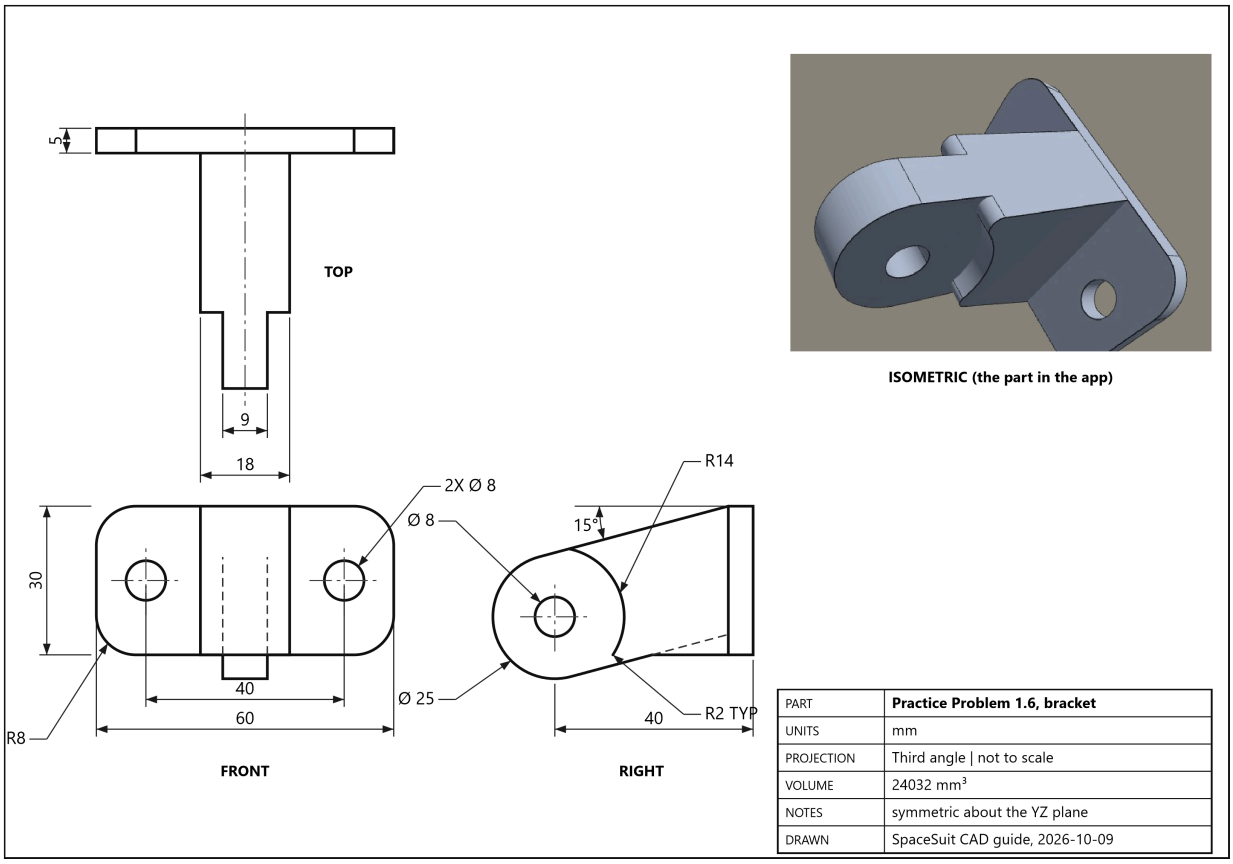


Fig. 4.1 The drawing: Practice Problem 1.6, bracket. Millimetres, third-angle projection, symmetric about the YZ plane.

> **TYPING COORDINATES AND SUMS**
>
> When the keypad asks for a number you can type a sum, such as **30/2** or **25/2**, and OK keeps the result. The Point tool asks for X and Y from where you pinched, so points land exactly where the drawing puts them.

1 · A new part, the plate's sketch on XZ

PARTS › New (check the unit key reads **mm**). Press SKETCH: an empty part opens on XY. On the board, EDIT › Plane steps the empty sketch to **XZ**, which is the plate's back face.

2 · The plate: 60 by 30 with R8 corners

Open Rect ▾ and choose C.Rect. Pinch the origin, point outwards, type **60**, OK, **30**, OK, then pinch to place it. EDIT › Fillet: pinch each corner (each rounds at 5 mm). CONSTRAIN › Dimension: pinch each round, type **8**, OK.

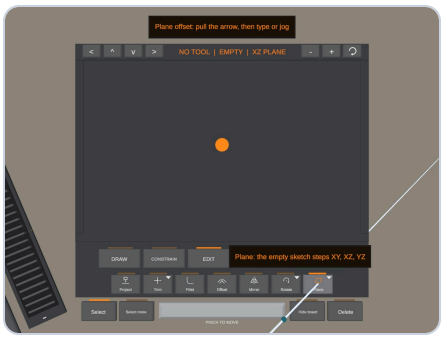


Fig. 4.2 An empty sketch on XZ.

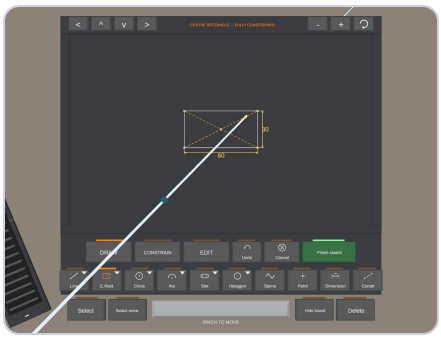


Fig. 4.3 The centre rectangle, 60 × 30.

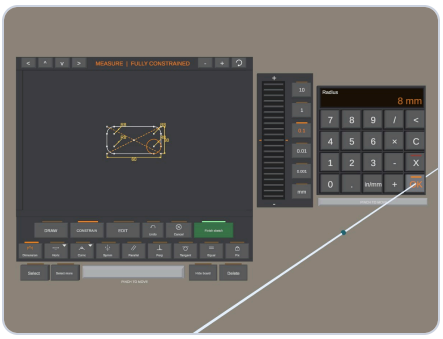


Fig. 4.4 R8 on each corner: fully constrained.

3 · The holes, 40 apart; extrude 5

Point: pinch the origin and type X **20**, Y **0**. Circle on that point: **8/2**. Do the same at X **−20**. Press Finish sketch, then FEATURE › CREATE › Extrude. Pull the arrow a little and type **5** (it goes towards you).



Fig. 4.5 Two Ø8 holes, 20 either side of the origin.

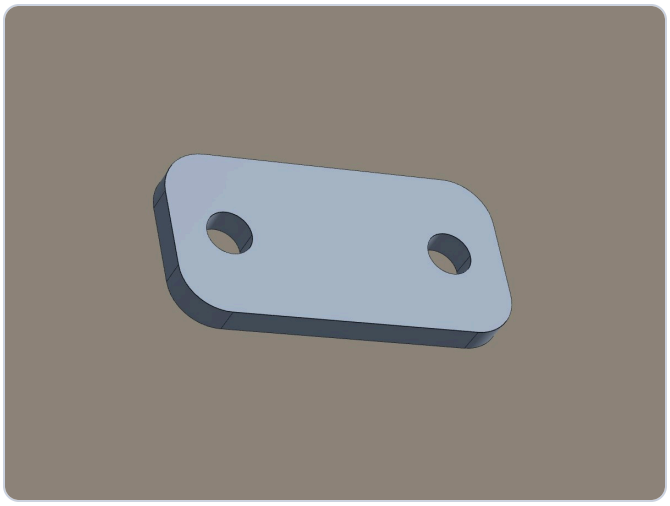


Fig. 4.6 The plate, 5 thick.

4 · The arm's sketch on YZ

Turn the part so you can see the YZ plane beside it. Press **SKETCH** and pinch the plane where nothing of the part is in front. Zoom in on the arm's place with **+** on the board's rim (the arrows pan).

Point from the origin: X **−5** (the plate's thickness), Y **30/2**: the plate's top front corner. Again: X **−40**, Y **0**, which marks the round end's centre line, 40 from the back face. Draw a short **Line** straight down from that point and make it construction with **Constr**.



Fig. 4.7 A sketch on YZ, zoomed in on the arm's place.

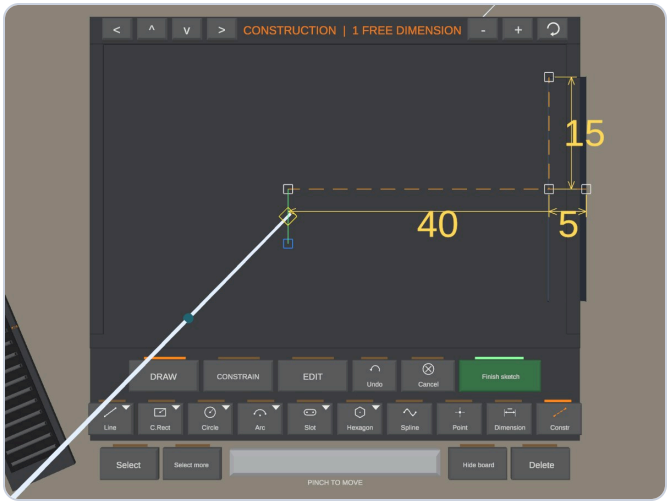


Fig. 4.8 The corner, and the construction line 40 from the back.

5 · The plate's face, the round end and its hole

Line from the corner straight down: type **30**, tap the display, then type **90** for the angle, OK, and pinch. This is the plate's front face. **Circle** on the construction line's end: **25/2**; again on the same point: **8/2**.

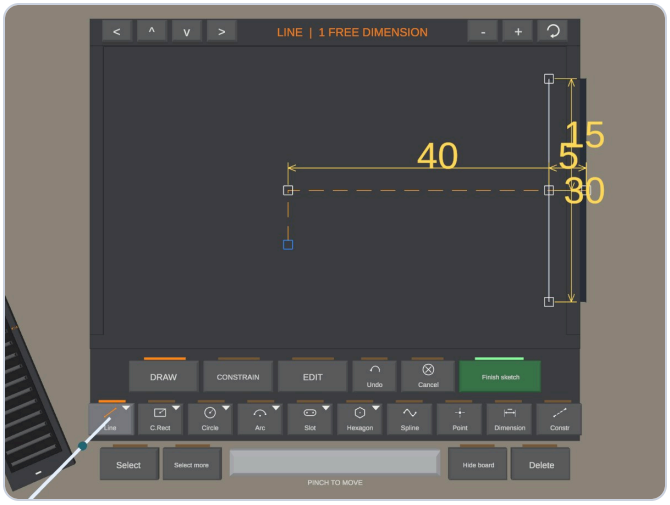


Fig. 4.9 The plate's face, 30.



Fig. 4.10 Ø25 and Ø8 on the line's end.

6 · The top edge at 15°, the bottom edge parallel

Line from the plate's corner to the end's rim, up and to the left of its centre. CONSTRAIN › Tangent: the line, then the circle. Dimension: the line, tap the display, **15**. Then a Line from the rim below the centre to the plate's face; make it Tangent to the circle and Parallel to the top edge.

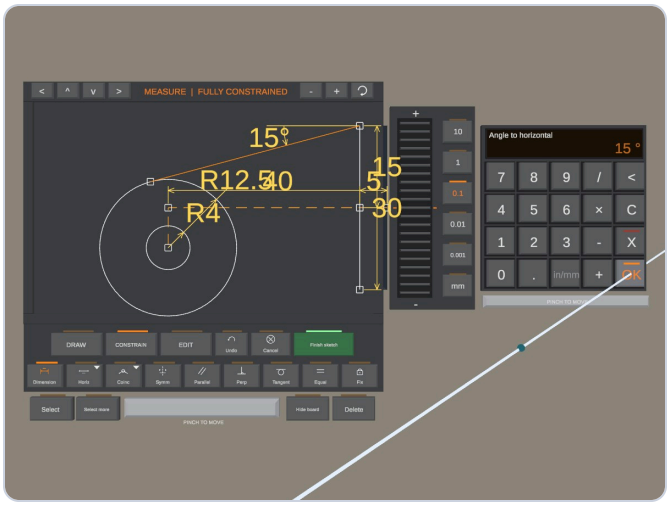


Fig. 4.11 Tangent, and 15° to horizontal.

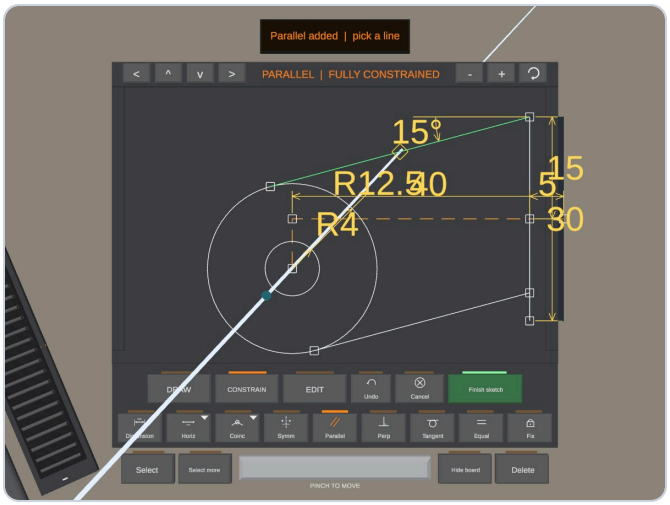


Fig. 4.12 The bottom edge: tangent and parallel.

7 · R14 round the end, the plate's bottom carried to it

Circle on the end's centre: **14**. A Line from the face's bottom end to the R14 (it holds on the circle), then Horiz. The board reads **FULLY CONSTRAINED**.



Fig. 4.13 R14 and the bottom line: fully constrained.

8 · The lug: 9 wide, Mid plane

Finish sketch, then FEATURE › CREATE › Extrude. The sketch makes several regions, and the readout counts the ones in use. Pinch the thin ring round the end, outside its two edges, to leave it out. Press Extent once for **Mid plane**, so the lug grows equally both ways. Pull the arrow a little and type **9**, OK.

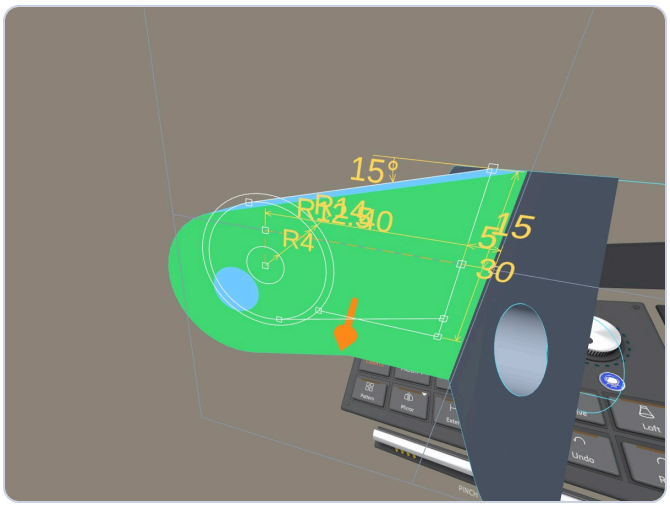


Fig. 4.14 The regions in use (green); the thin ring pinched out.

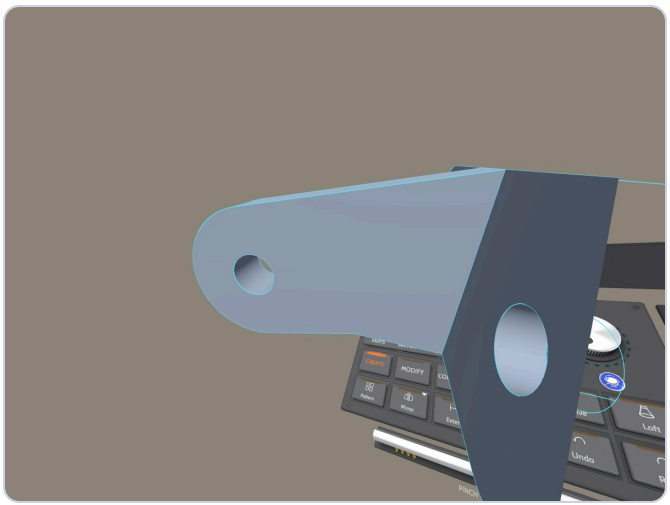


Fig. 4.15 The lug, 9 wide.

9 · The block: 18 wide, Mid plane

Tap the arm's sketch in the timeline and press Extrude again. This time pinch out the round end, the thin ring, the band between the end and the R14, and the band's bit under the plate's bottom, until **2 of 7 regions** are left: the arm by the plate and the bit under its bottom edge. Extent: **Mid plane**; type **18**.

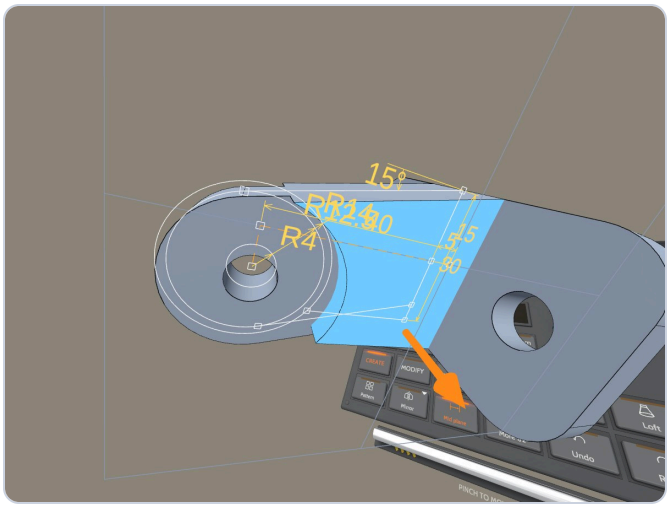


Fig. 4.16 Two regions left: the arm by the plate.

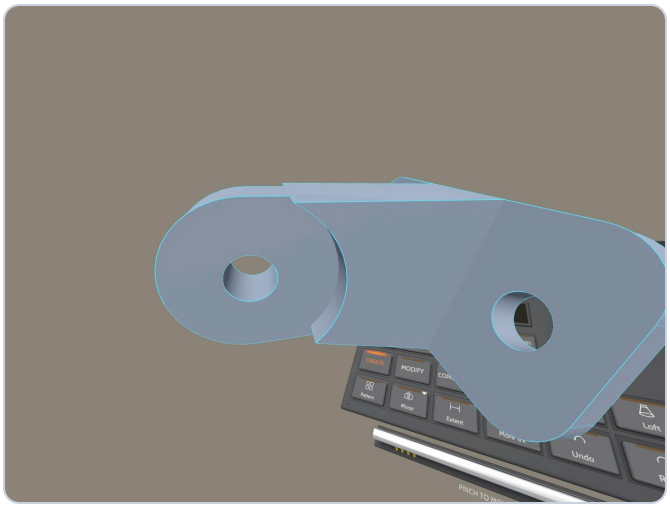


Fig. 4.17 The block, 18 wide.

10 · R2 on the block's four edges along the R14

FEATURE › MODIFY › Fillet. Grab the part and bring it up, clear of the console. Pinch the four edges where the R14 meets the block's top and bottom, two on each side of the lug, turning the part to see each one. Type **2**, OK. The bracket is done; PARTS › Export writes it as STEP and STL.

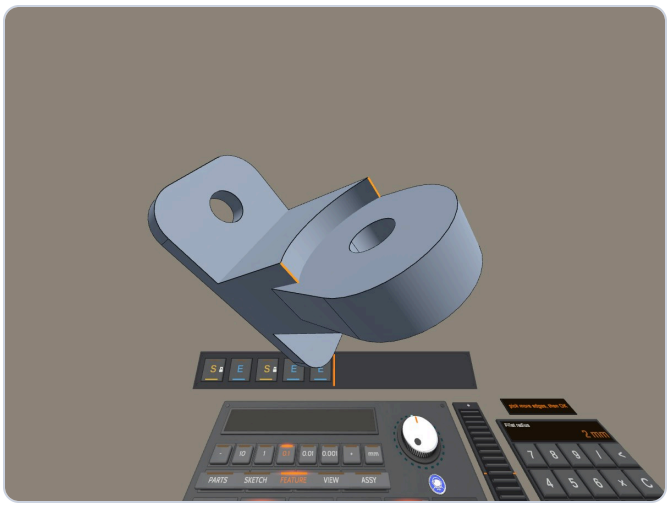


Fig. 4.18 Four edges picked (amber).

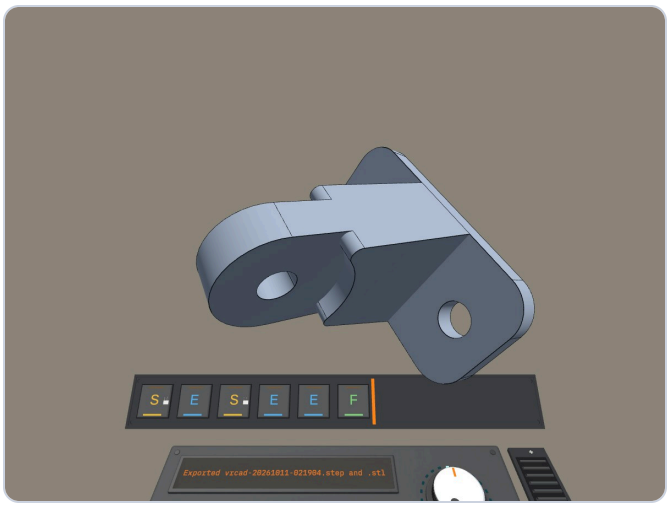


Fig. 4.19 Practice Problem 1.6, finished.

WHAT THIS PART TAUGHT

Points placed by typed coordinates, sums on the keypad, a sketch fully constrained with tangents, an angle and a parallel, one sketch used by two extrudes with different regions, Mid plane, and fillets on picked edges.

5 Quick start: an assembly

Put a pin through a plate in a few presses.

Assemblies are built from saved parts. Make the parts first in PARTS (each New starts another part and keeps the last one): here, a 60 × 40 × 10 plate with a Ø10 hole, and a Ø10 × 30 pin with a Ø16 × 4 head.

1. ASSY › New. Press Insert and choose each part. The first part is fixed in place; the others are free.
2. Open Mate ▾ and choose Coincident. Pick the underside of the pin's head, then the top of the plate. The readout says `Coincident1: solved, 3 free`.
3. Choose Concentric. Pick the pin's round face, then the hole. The pin drops into the hole.
4. Pinch a free part and move it: the parts move as far as their mates let them, so you can try a mechanism by hand.

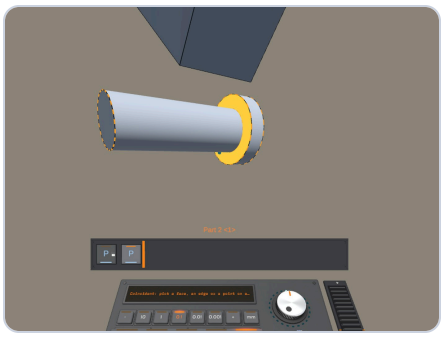


Fig. 5.1 Coincident: the head's underside picked (amber).

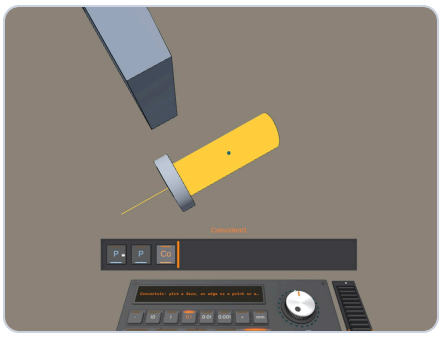


Fig. 5.2 Concentric: the pin's shaft picked.



Fig. 5.3 The pin in the plate.

The timeline lists the parts and the mates. Select a mate to change it with Value, Flip, Suppress or Delete. Select a part and press Edit part to change it in FEATURE: the assembly picks up the change when you come back.

Changing your mind

To	Do this
Change how a feature was made	Tap it in the timeline, press Edit (MODIFY) or **B**. Its own flow runs again with its values filled in; press Edit again to finish. The whole edit is one undo step.
Change just a size	Select the feature and press Value, or turn the dial. On a selected sketch, Value steps through its dimensions.
See an earlier stage	Drag the amber marker in the timeline to the left. New features go in at the marker.
Make a variant	PARTS › Config › New, then change values; switch between configurations with their keys.
Keep a copy	PARTS › Copy (Part 3 becomes Part 3B). Open lists your saved parts.
Undo	Undo / Redo, in every mode. The history survives a restart.

6 Cheat sheet

Where to find things.

I want to…	Press
Start a part	PARTS › New
Open a saved part	PARTS › Open
Bring in a STEP or drawing	PARTS › Import
Send the part out	PARTS › Export (STEP + STL)
Give it a material and mass	PARTS › Material
Sketch	SKETCH, then pick a plane or face
Extrude from nothing	FEATURE › Extrude starts a sketch for you
Cut	CREATE › Extrude ▾ Cut
Cut all the way through	CREATE › Extent → Thru all
Revolve, loft, sweep, rib	CREATE
A standard hole	CREATE › Hole
Repeat or mirror features	CREATE › Pattern / Mirror

I want to…	Press
Round or bevel edges	MODIFY › Fillet ▾ (Chamfer, Full round)
Hollow it out	MODIFY › Shell
Taper faces	MODIFY › Draft
Move a body	MODIFY › Move
Change a feature	MODIFY › Edit (or B)
A reference plane or axis	FEATURE › CONSTRUCT
Measure	VIEW › Measure
Do sums	VIEW › Calc
See it at real size	VIEW › 1:1, then Lock
See the timeline as a tree	VIEW › View
Build an assembly	ASSY › New, Insert, Mate
Licences and version	VIEW › About

GOOD HABITS

Read the readout: it always says what is selected and what the app expects next. Dimension sketches before extruding them. Pick a face or a plane before you draw, rather than empty space. If something goes wrong, Undo.

THIS BUILD

Much of this build is new and has been checked in the desktop preview but not yet on the headset; details may still change. The full user manual covers the earlier features in more depth.



SpaceSuit CAD

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